RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6701 RF and Microwave Engineering
Name of the Faculty member: Mrs.V.SrirengaNachiyar
Name of the Topic: Amplifier Power Relations
Name of the Innovative Practice: Mind map
Date & Time: 14.08.2019 & 20 Minutes
ICT tool used: Whiteboard

Description:

Mind Map Technique:

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Learning Outcomes:

1. The students will be able to differentiate the operating power gain, Available power gain, Transducer gain.
2. The students can be acquired better knowledge about the amplifier power relations.
3. The students can be easily identified the corresponding amplifier power relation equations for solving the problems.

Use of appropriate method:

Justification for choosing the Mind Map Activity:

Mind map activity gives the graphical representation of Amplifier power relations. It differentiates the Power gain, Available power gain and Transducer Power gain clearly which will be helpful for understanding the concepts of amplifier power relations to the students.

Effective presentation:

I have planned the activity for 15 minutes but students don’t know how to do this activity. So, I have explained how to draw the mind map for this particular topic and also the benefits of mind map during examination. I encouraged the students which it takes another 5 minutes for completing the activity.

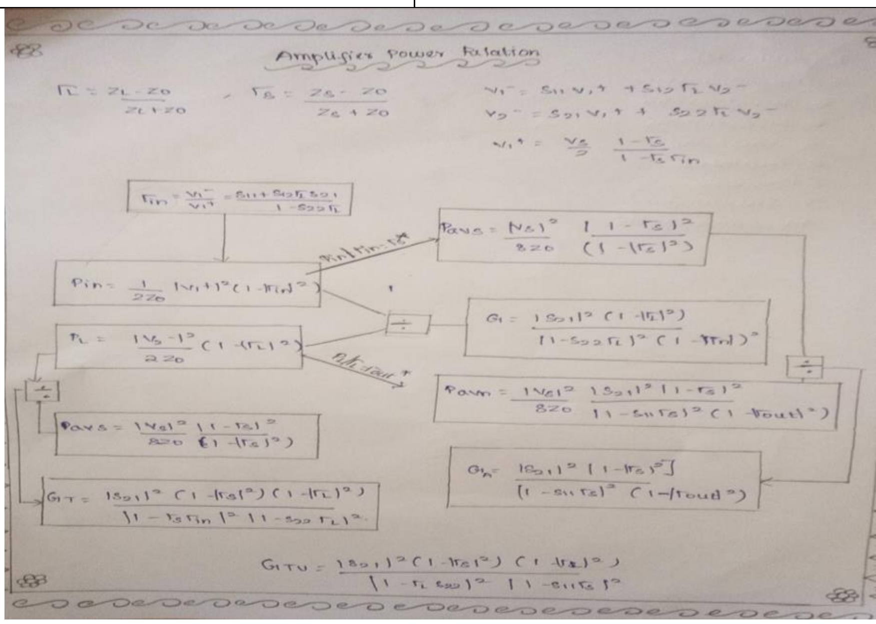
Reflective Critique:

<u>Challenges:</u>

The students confused about the different types of amplifier power gains (Operating Power gain, Available Power Gain and Transducer Power gain) which was derived in the class room. After that, I have chosen Mind Map as an Innovative practice method for better understanding about Amplifier power relations.

<u>Benefits:</u>

The Students eagerly participated in that activity and then they acquired the concept of amplifier power relations. Due to this activity, the students were able to recall the topic and remember the important terms easily.

Innovative Teaching Method Execution	
Amplifier Power Relations:– Mind Map	
	
	

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	2	1					2	1		

CO2: The students will be able to formulate solutions for stability and impedance matching issues in microwave amplifiers

References:

1. https://www.youtube.com/watch?v=V9S6o4tcc_I
2. Reinhold Ludwig and Gene Bogdanov, “RF Circuit Design: Theory and Applications

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Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6701 RF and Microwave Engineering
Name of the Faculty member: Mrs.V.SrirengaNachiyar
Name of the Topic: Circulator and Isolator
Name of the Innovative Practice: Role Play
Date & Time: 14.08.2019 & 15 Minutes
ICT tool used: Whiteboard

<u>Description:</u>
Role Play Technique:

Role-play is a technique that allows students to explore realistic situations by interacting with other people in a managed way in order to develop experience and trial different strategies in a supported environment.

Role play is an educational technique in which people spontaneously act-out problems of human relations and analyze the enactment with the help of other role players and observers.

Role playing is effective when the topic involves person to person communication or interactions. It can allow everyone to participate. The overall mood a training session can be improved by the excitement that role playing provide.

<u>Learning Outcomes:</u>

1. The students can be acquired the better knowledge about circulator and isolator.
2. The students can be easily differentiated the working principles of 3 port and 4 port circulators and 2 port isolators.\

<u>Use of appropriate method:</u>
Justification for choosing the Role Play Activity:

Role play activity gives the detailed explanation of the circulator and isolator. It differentiates the working of Circulator and Isolator and its S-parameters clearly which will be helpful for understanding the concepts of the device.

<u>Effective presentation:</u>

I have planned the activity for 15 minutes but students don't know how to do this activity. So, I have explained the activity and also the benefits of doing this activity for this particular topic.

Reflective Critique:

<u>Challenges:</u>

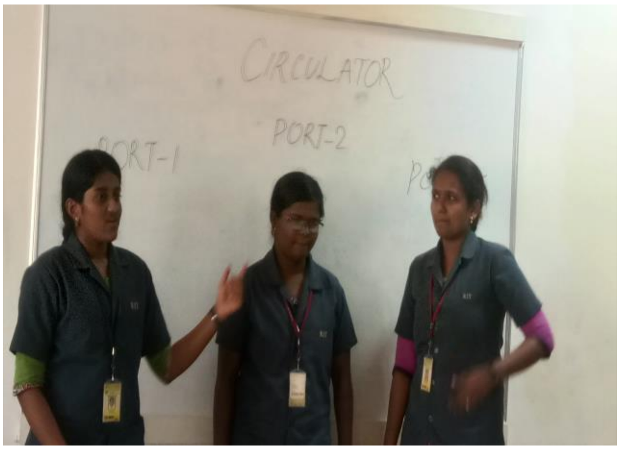
The students confused about the working of Circulator and Isolator which was discussed during class hours. After that, I have chosen Role play as an Innovative practice method for better understanding about working principle.

Benefits:

The Students eagerly participated in that activity and then they acquired the concept of Circulator and Isolator. Due to this activity, the students were able to recall the topic and remember the important terms easily.

Innovative Teaching Method Execution

Circulator, Isolator – Role Play



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	2	1						2	1		

CO3: The students will be able to explain the basic concept of various microwave devices

References:

1. MIT opencourseware
2. Annapurna Das and Sisir K Das, "Microwave Engineering"

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Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6701 RF and Microwave Engineering
Name of the Faculty member: Mrs.V.SrirengaNachiyar
Name of the Topic: Travelling Wave Tube amplifier
Name of the Innovative Practice: Brain storming
Date & Time: 17.09.2019 & 15 Minutes

Description:

Brainstorming Technique:

Brainstorming is a method for generating ideas to solve a design problem. It usually involves a group, under the direction of a facilitator. The strength of brainstorming is the potential participants have in drawing associations between their ideas in a free-thinking environment, thereby broadening the solution space. The purpose of this is to gain as many ideas as possible for later analysis. Out of the many ideas suggested there will be some of great value. Because of the free-thinking environment, the session will help promote radical new ideas which break free from normal ways of thinking.

Learning Outcomes:

1. The students will be able to remember the important terms and definitions
2. The students can easily understand the working principle of Travelling Wave Tube amplifier.

Use of appropriate method:

Justification for choosing Brainstorming Activity:

Before the topic of Traveling wave tube amplifier, I have taught two cavity Klystron amplifier and Reflex Klystron oscillator. The working principle of TWT and other klystron amplifiers are same. Therefore, students should know about microwave tubes. For that reason, I have chosen Brainstorming activity for this topic. Definitely, it helps to understand the working principle of TWT.

Effective presentation:

I have planned the activity for 10 minutes but students struggled to recall about microwave tubes and then I gave another 5 minutes to recall it. Totally, it took 15 minutes to complete it.

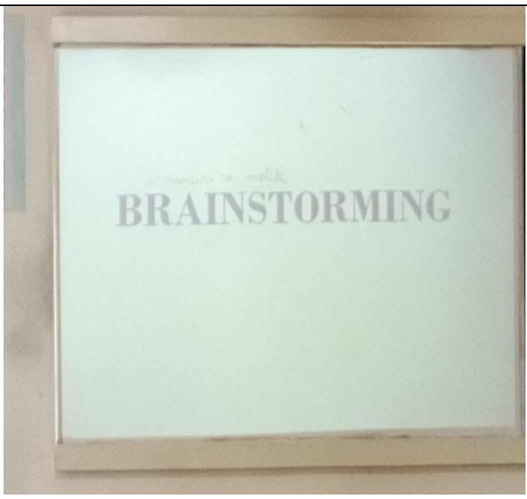
Reflective Critique:

Challenges:

Some of the students couldn’t able to recall the concept while doing the activity. It takes more time to recall about microwave tubes which was taught in previous lectures.

Benefits:

After that, students eagerly participated and brainstormed their views about travelling tube amplifier. At last, I concluded about the working principle and operation of travelling tube amplifier.

Innovative Teaching Method Execution	
Travelling Wave Tube amplifier- Brainstorming	
	

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	2	2				2	2	2		2

CO4: The students will be able to elaborate the concepts involved in Microwave generation

References:

1. Annapurna Das and Sisir K Das, "Microwave Engineering"
2. https://www.google.com/search?safe=strict&ei=WOjkXffiBv-V4-EPvo6UiAQ&q=importance+of+brainstorming+pdf&oq=importance+of+brain+stor&gs_l=psy-ab.3.0.0i10l10.545310.547992..550114...0.3..0.111.1168.8j4......0....1..gws-wiz.......0i71j0i67j0j0i273j0i131.TdXTdxWi6RY

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Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6701 RF and Microwave Engineering
Name of the Faculty member: Mrs.V.SrirengaNachiyar
Name of the Topic: Measurement of Impedance, Frequency, Power
Name of the Innovative Practice: Demonstration
Date & Time: 28.09.2019 & 20 Minutes

Description:
Demonstration:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

Learning Outcomes:

1. The students can be acquired better knowledge about the measurement of Impedance, Frequency using Slotted Line Section.
2. The students can be easily operated the Slotted Line Section for measuring the indirect way of frequency.

Use of appropriate method:

Justification for choosing Brainstorming Activity:

All the students did measurement of Frequency and Power using slotted line section and VSWR meter in the Optical and Microwave laboratory. Likewise, the students want to do the experiment to measure the impedance using slotted line section and VSWR meter. For that, I want to demonstrate the slotted line section experiment for measuring Impedance, Frequency, and Power.

Effective presentation:

I have planned the Slotted line section experiment demonstration nearly 20 minutes.

Reflective Critique:

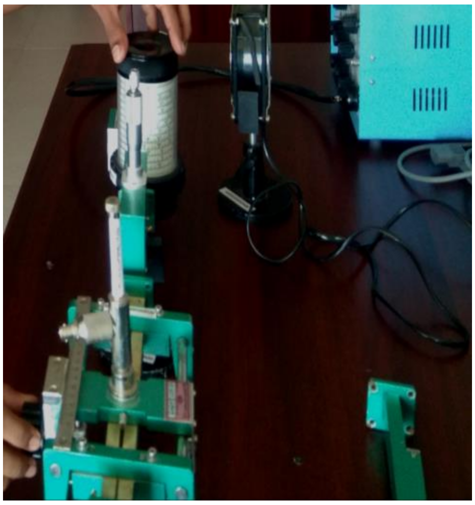
Challenges:

During the class, most of the students confused about measuring the impedance, frequency and power using Slotted line section and VSWR meter. After demonstration, they can be acquired better knowledge about measuring these parameters. Some of the students not shown interest during the demonstration

Benefits:

then I told the importance of hardware demonstration and correlation between theory and laboratory courses. Then students enjoyed doing the experiment.

Innovative Teaching Method Execution
Measurement of Impedance, Frequency, Power - Demonstration



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	3	3	1					2	1		

CO5: The students will be able to measure and analyse Microwave signal and its parameters.

References:

1. Annapurna Das and Sisir K Das, “Microwave Engineering”
2. Reinhold Ludwig and Gene Bogdanov, “RF Circuit Design: Theory and Applications.